Opportunistic Target Selection

Early Directional Commitment for Query-Efficient Black-Box Adversarial Attacks

Florent Tariolle[1] Florian Yger[1,2]

[1]INSA Rouen Normandy [2]LITIS

1. Motivation: Realistic Black-Box Attacks

Why black-box?

- Real attackers rarely know the model weights, architecture, or gradients.
- Many deployed systems still expose scores, confidence values, or ranked labels.
- Query-efficient score-based attacks are therefore the closest setting to a practical threat model.

Score-based black-box attacks see probabilities, not gradients. Standard untargeted losses mainly lower the true-class confidence:

$$\mathcal{L}_{\text{untarg}}(x', y) = P(y \mid x') \quad \text{or} \quad \log P(y \mid x').$$

- Probability mass is freed, but no non-true class is chosen.
- The trajectory can drift across several class basins.
- Query budget is wasted before misclassification.

Goal. Pick a viable target from the trajectory, then commit early to avoid drift.

Gap.

- Targeted attacks need a target before the run starts.
- Untargeted CE/probability losses do not decide which competitor should take over.

2. Method: Opportunistic Target Selection

OTS keeps the proposal mechanism unchanged. Only the loss schedule changes after a short untargeted prefix.

Algorithm.

1. Run the original untargeted attack for T iterations.
2. Lock onto the leading non-true class
$$t = \arg\max_{k \neq y} P(k \mid x'_T).$$
3. Continue the same attack, but maximize the target confidence $P(t \mid x')$.
4. Stop when the classifier prediction is no longer y.

- Irreversible lock: no repeated exploration.
- Target choice matters: random targets fail; the leading class after a short prefix is viable.
- Stable for $T \in \{5, \ldots, 500\}$: rankings appear early.

3. Experimental Protocol

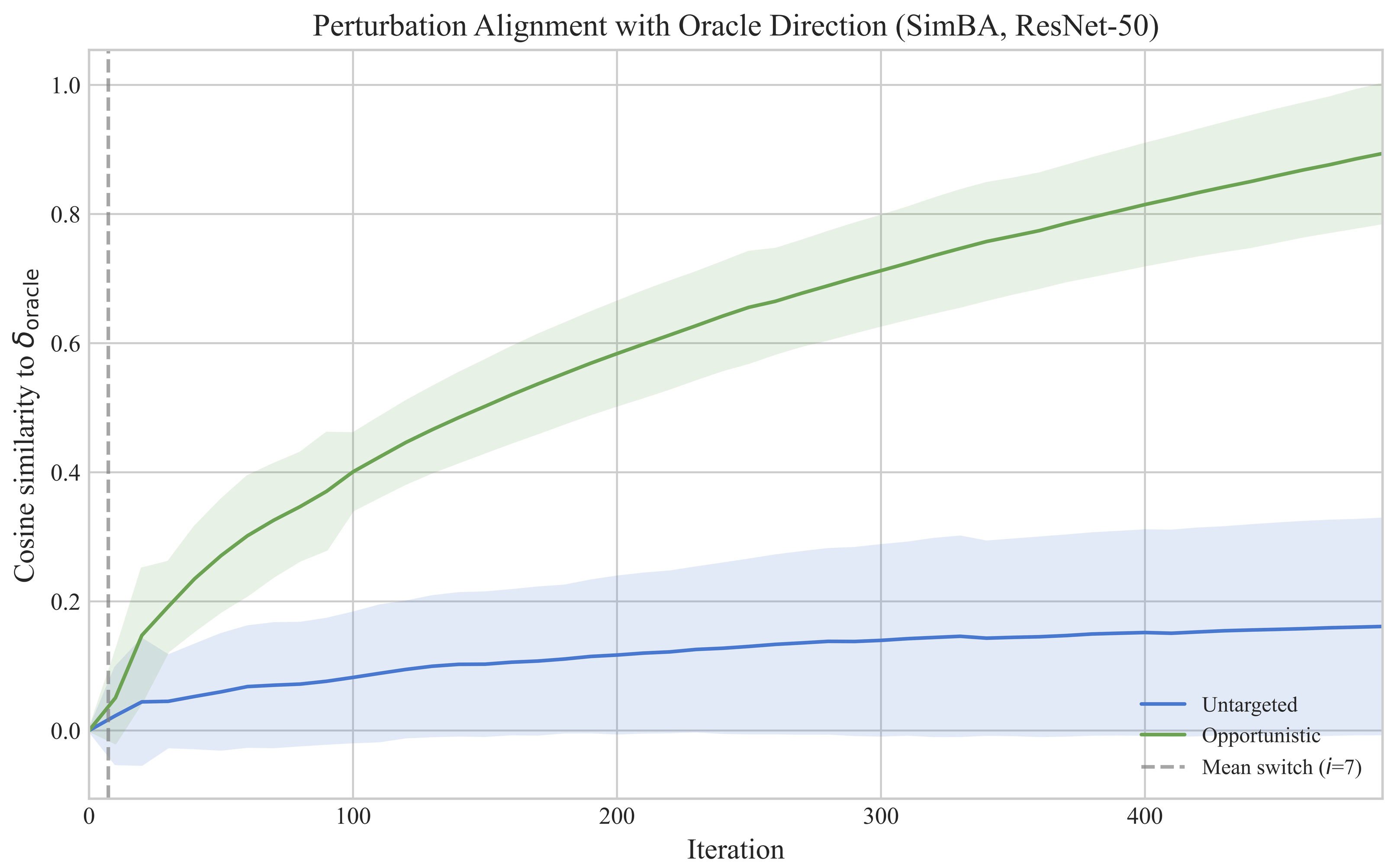
Models	AlexNet, ResNet-18, VGG-16, ResNet-50, ViT-B/16
Attacks	SimBA, Square Attack with CE loss, Bandits
Modes	Untargeted baseline, targeted oracle, OTS
Budget	$\epsilon = 8/255$ in L_∞, 10K iterations
Scale	100 ImageNet images per model, 4,500 standard-model runs
Metrics	Success rate and censored mean iterations

Oracle target: same-seed untargeted endpoint, used as a trajectory-specific ceiling.

4. When and Why OTS Helps

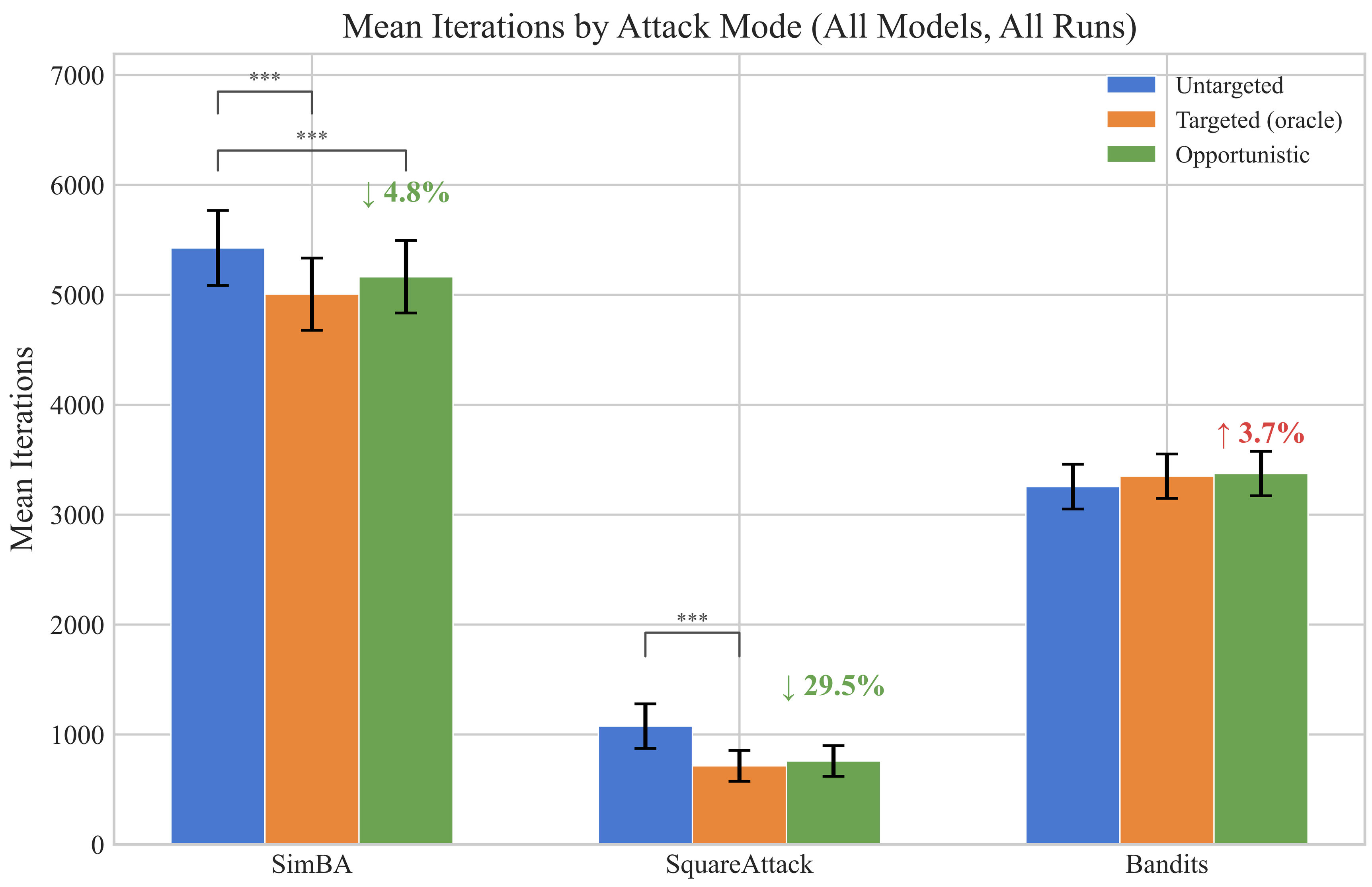
Attack	Direction source	Expected OTS effect
SimBA	No target tracking	Large gain from early commitment
Square CE	y-only CE	Margin-like gain
Bandits	Gradient estimate	Small or negative gain

- Helps when the base loss drifts.
- Can hurt when the base attack already has a direction.



- Mechanism: OTS turns early score information into a fixed direction.
- Fixed t mimics the margin competitor:
$$\mathcal{L}_{\text{margin}} = f_y(x') - \max_{k \neq y} f_k(x').$$

5. Main Results



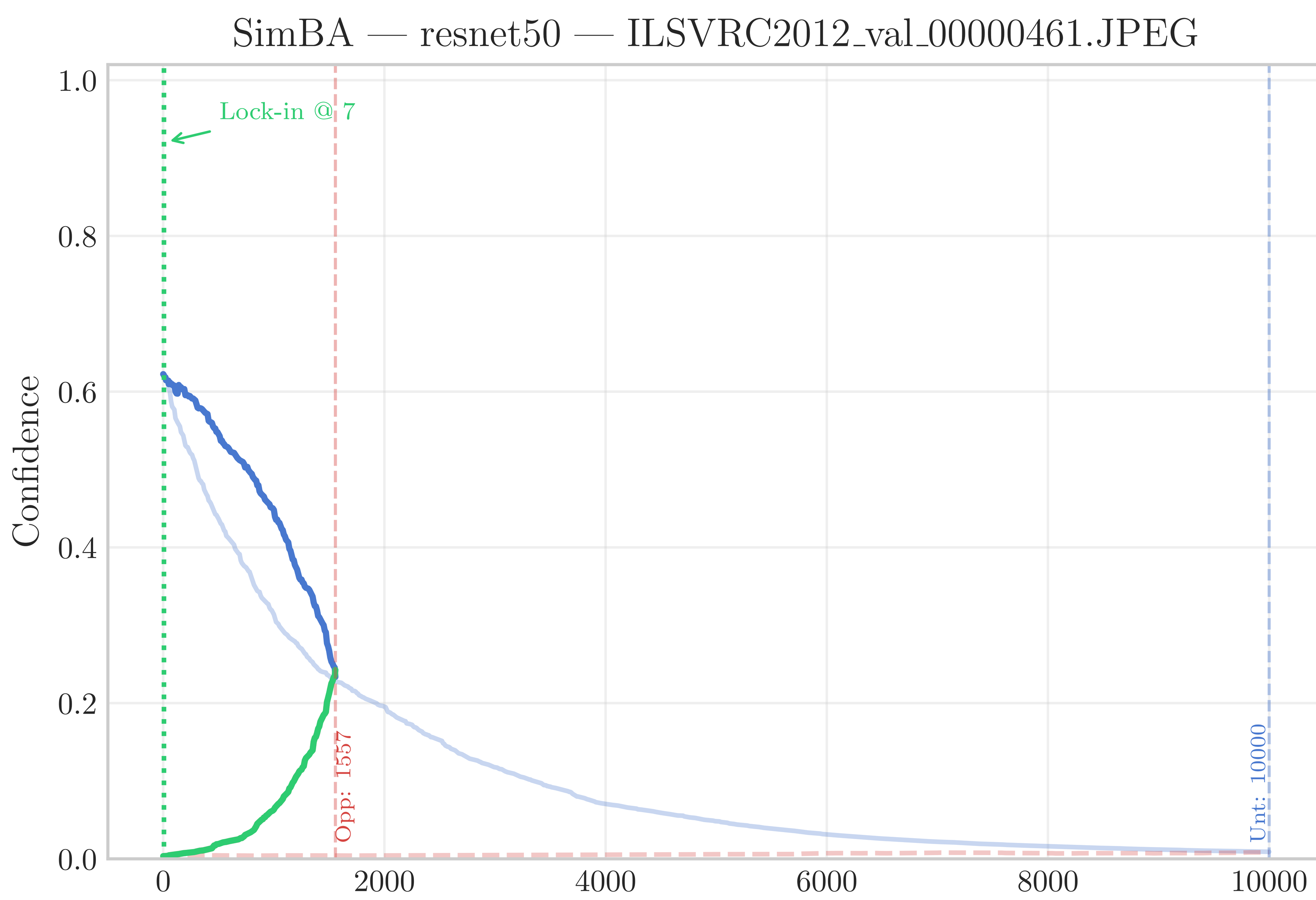
Attack	Untargeted	OTS	Oracle
SimBA	67%	**75%**	76%
Square CE	96%	**99%**	99%
Bandits	37%	34%	35%

+27 pp	**43%**	**0**
SimBA success on ResNet-50	relative iteration reduction	gradient access

Main finding.

- Near-oracle behavior for SimBA and Square CE.
- Bandits gets worse: it already follows an estimated gradient.

6. Lock-In Dynamics



- Untargeted: suppresses the true class, then drifts.
- OTS: locks class 988 and keeps increasing it.
- Only a stable competitive target is needed.

7. Limits and Takeaways

Take-away.

- Explore briefly, then lock the strongest non-true class.
- The gain comes from avoiding class drift under score-only feedback.
- OTS is useful when the base attack lacks a stable direction; it can hurt when the attack already follows one.

Current limits.

- Robust ImageNet models: flat success across variants.
- No medium-difficulty regime where commitment pays off.

Future work.

- CIFAR-10, L_2 budgets, randomized defenses.
- Adaptive re-locking if the target drops in rank.

Acknowledgments.

- LITIS funded the expenses for CAp 2026.
- MesoNet (CRIANN, Rouen) provided the A100 GPU used for the benchmarks.